# RASHEDUL ISLAM

**Address:** Cox's Bazar Sadar, 4700, Bangladesh
**Personal portfolio:** rashed-mecha.netlify.app **Email:** rashed.me.82@gmail.com **LinkedIn:** linkedin.com/in/rashedown
**Mobile:** (+880) 1855 362 882

## Overview

A highly driven Mechanical Engineering researcher and prospective PhD student specializing in Computational Fluid Dynamics (CFD) of biofuels, hydrogen combustion, and renewable energy structure. Backed by peer-reviewed international journal publications and credentials as a Certified SolidWorks Expert (CSWE), I leverage high-fidelity multiphysics modeling and advanced CAD to solve complex thermal-fluid, energy storage, molecular dynamics and clean energy engineering challenges.

## Education

- **Hajee Mohammad Danesh Science & Technology University (HSTU)** Dinajpur, Bangladesh
  *B.Sc. in Mechanical Engineering; CGPA: **3.363 / 4.00*** *Jan 2019 – Dec 2022*
  - **Dissertation**: Investigate the Performance of Hydrogen Fuel and Compare with Conventional Fuels in Port Fuel Injection SI Engine using CONVERGE CFD. *(Output*: Research paper was published in a Q1 Journal- Next Energy, 2025)
  - **Supervisor**: Md. Rasel Ahmed
  - **Synopsis**: This dissertation presents the exciting frontier of hydrogen as a clean, high-efficiency fuel in spark-ignition engines, using advanced Converge CFD simulations to compare it against conventional fuels like gasoline, methane, and M10. Through a series of simulated combustion trials, hydrogen proves itself not just as a viable alternative but a standout performer-delivering smoother cylinder pressures, higher thermal efficiency, and near-zero emissions at optimal conditions. Particularly at an equivalence ratio of 0.4 and 3000 rpm, hydrogen showcases its potential to power the future of transportation with minimal environmental impact. The research boldly positions hydrogen as a clean-energy champion ready to replace carbon-heavy fuels in a world urgently seeking sustainability.

## Experience

- **HSTU and RUET**
  *Research Assistant (RA) (PI: Md. Rasel Ahmed)* *Aug 2024- On going*
  - **Data collection**: Visited Wind Turbine Powerplant at Cox's Bazar and collected the data for the study.
  - **Model development and validation**: Biogas Generator, solar panel and wind turbine arranges the best outcome.
  - **Analysis and writing**: Analyzed the off-grid power cases using HOMER Pro and plotted the results.

- **Placid Bangladesh (Intern)**
  *Mechanical Engineering Intern* *Jan 2025 – May 2025*
  - Assessed mechanical systems and completed product optimization resulting in reduced material waste and improved operational efficiency.
  - Participated in structured training sessions on technical operations, maintenance procedures, and quality standards.
  - Gained practical exposure to maintenance scheduling, equipment inspection, and corrective action follow-up in a field environment.

## Research Interests

My work centers on computational fluid dynamics and hydrogen combustion, extending into materials science, heat transfer, and additive manufacturing. Renewable energy systems- particularly hybrid, off-grid optimized solutions- round out this focus, with simulation-driven modeling as the common thread across all of it.

## Skills Summary

- **CFD, Simulation & CAD Tools:** CONVERGE CFD, COMSOL Multiphysics, SolidWorks, AutoCAD, Solid Edge
- **Energy Modelling & Analysis:** HOMER Pro, PVsyst, RETscreen, Decision Science, Hybrid System Design & Optimization (NSGA-II), LCOE / NPC / LCOH Analysis, Load Assessment, LCA & Uncertainty Analysis (Montecarlo)
- **Atomistics & Molecular Dynamics:** LAMMPS, OVITO
- **Technical Documentation & Capacity Building:** Energy Performance Reports (MS Office and Latex), System Diagrams, Maintenance Records & Feasibility Studies, Power Assessments, Training & Technician Coaching, Hands-on Field Instruction
- **Programming:** Python, MATLAB
- **Soft Skills:** Field Logistics Coordination, Team Leadership & Project Management

## Journal Publications/Under Peer-Review

- R. Islam, M. R. Ahmed, and S. I. Sayem, "Techno-Economic and Environmental Optimization of Advanced Hybrid Renewable Energy Systems with Green Hydrogen Production for Off-Grid Coastal Community Electrification in Bangladesh," *Energy Conversion and Management: X, p. 102179*, 2026, doi: https://doi.org/10.1016/j.ecmx.2026.102179
- S. M. Saklain Mostak Saumik, Naimur Rahman, Md Abdullah Al Noman, Rashedul ..., "Techno-economic investigation of a solar Photovoltaic (PV) energy model using PVsyst, Homer, RETscreen: a case study in Bangladesh," *Discover Electronics 2 (1), 81, 2025* , doi: https://doi.org/10.1007/s44291-025-00119-1
- R Islam, SMA Rahman, MR Islam, MR Islam, MR Ahmed, MRI Sarker, "Performance and emission analysis of hydrogen and conventional fuels in PFI SI engines using CONVERGE 3.0," *Next Energy 9, 100404, 2025*, doi: https://doi.org/10.1016/j.nxener.2025.100404
- [*Submitted · Journal*]- Comparative Molecular Dynamics Study of Mechanical, Thermal, and Fracture Behavior in Ti, Si, and Ti-Si-Doped Monolayer Graphene
- [*Under Review · Journal*]- Computational Thermo-Hydraulic Performance Evaluation of a Shell and Tube Heat Exchanger Using Concave Triangular Ribs
- [*Under Review · Journal*]- Exact Perron-Frobenius Orbits and Quantitative Convergence to Equilibrium for the Fully Chaotic Logistic Map
- [*Submitted · Review Paper*]- A Comprehensive Review of Techno-economic investigation and state-of-the-art Colored Hydrogen Production with Its Challenges
- [ *Submitted · Review Paper* ]- Hydrogen Storage Technologies: A Comparative Review of Materials, Performance Metrics, and Techno-

Economic Feasibility

- [ *Submitted · Conference* ]- A Comprehensive Investigation of MXene and Flow Capacitive Deionization for Sustainable Energy Storage Applications

## Undergraduate Projects

- **Electro-Mechanical Wheel Chair:** The goal of this project is to design and build an affordable solution to the mobility issues faced by those with physical limitations. Utilized SolidWorks for designing CAD models Pro to render images and animations. (2022)
- **Five Speed Gear Box Mechanism with High RPM, High Gear Ratio and Fuel Efficiency:** Used in coaxial input and output, more accuracy, high torque and high gear (6:1) ratio. Designed for improved power transmission efficiency, optimized gear mesh ratios, and fuel economy. (2020)

## Professional and Skill Trainings

- **5 days of Technical Training on Coal Testing at Coal Power Generation Company Bangladesh Limited (CPGCBL)**: Completed hands-on training at Matarbari Coal Power Plant, Cox's Bazar, conducted by Hunan Sundy Science and Technology Co., Ltd., covering coal quality analysis using Hardgrove Grindability Index Tester, Coulomb Sulfur Analyzer, Ash Fusion Tester, and Carbon, Hydrogen, and Nitrogen Analyzer. (2026)
- **28 days of Mechanical Workshop**: A 109-hour industrial mechanical training program covering workshop technology, metro-logy and drawing, bearings and lubrication, process plant machinery, refrigeration and A/C, power transmission and alignment, bench fitting, industrial ancillaries, plant safety and instrumentation, and including the Ashulia Gas Powerplant, Narsingdi; Bangladesh. (2023)
- **2 days workshop visit on Production Process of Machineries in Saidpur Railway Workshop**: Participated in scheduled training sessions and lab works conducted by railway mechanical experts, gaining hands-on experience with workshop projects focused on generation and power transmission , wheel and axle profiling using heavy machinery, major accident restorations, structural welding, industrial painting, and modern technology retrofitting to upgrade older trains and keep the railway network running smoothly and safely. (2019)
- **Field visit: Wind Turbine Power Plant, Cox's Bazar**: Operational energy data collection for off-grid hybrid energy research. (2024)

## Achievements and Awards

- Champion as a team in Idea Generation Contest which is a segment of “INNOVIS-2020”, IEEE | HSTU - October, 2020
- Bronze Medalist, Divisional Physics Olympiad, Bangladesh Physics Olympiad - August, 2016
- Bronze Medalist, Regional Mathematical Olympiad, Bangladesh Mathematical Olympiad - January, 2017

## Certifications

- (2026) Coal Testing and Laboratory Instrumentation (Hunan Sundy Science and Technology Co., Ltd., CPGCBL, Bangladesh)
- (2026) AI for Mechanical Engineers (University of Michigan)
  - AI for Design and Optimizaiton
  - AI for Autonomous Vehicles and Robotics
  - AI for Energy and Biomedical Applications
- (2026) Materials Science for Advanced Technological Applications (Arizona State University, USA)
  - Phase Diagrams
  - Hardening, Polymers, Properties
  - Electrical Properties and Semiconductors
  - Ceramics and Composites
- (2026) Materials Science: 10 Things Every Engineer Should Know (University of California, Davis)
- (2026) Energy Production Distribution & Safety (University of Buffalo and Stony Brook University, USA)
  - Electric Power System
  - Natural Gas
  - Safety in the Utility Industry
  - Energy: The Enterprise
- (2026) Renewable Energy Technology Fundamentals (University of Colorado Boulder, USA)
- (2026) MATLAB Simulink (MATLAB)
  - MATLAB Mathworks
- (2020) SolidWorks Certified Additive Manufacturing Associate, Certified CAD Professional, Certified Sheet Metal Professional (SolidWorks (Online))
- (2017) Certified on Physics Olympiad in Divisional Region (Bangladesh Physics Olympiad)
  - Physics Olympiad
- (2021) Workshop on Python PL (IEEE HSTU and University of MICHIGAN)
  - Python Basics
- (2025) Six Sigma Yellow Belt (SCRUMstudy)
- (2020) INNOVIS-2020 Idea Generation Contest - Champion Team (IEEE HSTU Student Branch, Bangladesh)

## Relevant Affiliations

- **Cox’s Bazar Youth Development Society (CYDS)**: Executive Member (2024 - 2025)
- **Institution of Mechanical Engineers (IMechE)**: Affiliated Member - Membership Id: 80778687 (2019 - 2024)
- **American Society of Mechanical Engineers (ASME)**: Student Member - Membership Id: 103515864 (2019 - 2024)

## Volunteer Experience

- **Robo-riot Contest Event Organizer, Deceleration - A National Mechanical Festival** *2022 – 2023*
  *Organized and conducted two segments of the national inter-university mechanical festival at HSTU*
- **Autodesk Ambassador Hub** *2019 – 2021*
  *Shared news about Autodesk products, events, and completed challenges on social media platforms*
- **Adjunct Teacher, Cox's Bazar Govt. High School** *2017 – 2018*
  *Conducted weekly classes for junior categories*

## References

**Md. Rasel Ahmed**
Lecturer
Department of Mechanical Engineering, RUET
Email: rasel@me.ruet.ac.bd
Mobile: +880 1776 976 345

**KH. Nazmul Ahshan**
Chairman and Assistant Professor
Department of Mechanical Engineering, HSTU
Email: nazmul.me@hstu.ac.bd
Mobile: +880 1884 598 886